

SOFTWARE ENGINEERING

Lab 1: Requirements Engineering & UML Use-Case Modelling

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Problem Statement 65: Urban Rooftop Gardening Co-op App

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow gardeners to log seasonal crop planting dates, computing estimated harvest windows and expected yield weight.	High	Pass: Harvest alert generated within predicted window. Fail: Invalid planting date sequence accepted.	Core function of the co-op app; drives harvest planning and yield tracking for members.
FR-002	Functional	The system shall allow a Co-op Gardener to register a rooftop plot with location, size, sunlight exposure, and soil type.	High	Pass: Plot appears on the gardener's dashboard with a unique Plot ID immediately after submission. Fail: A duplicate plot (same location) is accepted without warning.	Establishes the base record every other feature (logging, trading, approval) depends on.
FR-003	Functional	The system shall allow Co-op Gardeners to	Medium	Pass: A posted tip is visible to other members	Supports the peer knowledge-sharing

		post and browse soil maintenance tips visible to all co-op members.		within the same session. Fail: A submitted tip is not saved or not shown to other members.	goal central to the co-op's community mission.
FR-004	Functional	The system shall allow a Co-op Gardener to list surplus produce for trade with quantity, unit, and pickup window.	High	Pass: Listing appears on the shared Trade Board immediately with the correct quantity and pickup window. Fail: A listing with zero or negative quantity is accepted.	Core value proposition of the app — enabling produce exchange between members.
FR-005	Functional	The system shall require a Community Leader to review and approve a new rooftop plot registration before it is visible to other members.	Medium	Pass: An unapproved plot is not visible to other members until the leader approves it. Fail: An unapproved plot is publicly visible.	Maintains data quality and prevents fraudulent or duplicate plot claims in the community.
NFR-001	Non functional (Performance & Security)	The produce trade board must update available surplus listings in	High	Pass: Benchmarking tests confirm target latency and security standards	Real-time visibility is essential so surplus produce is claimed

		real time across community members.		under simulated peak load.	before it spoils.
NFR-002	Non functional (Reliability & Availability)	The app shall remain usable with intermittent network connectivity; data entered offline (e.g., logging a harvest on the rooftop) shall sync automatically within 5 minutes of reconnecting.	Medium	Pass: Offline log entries sync automatically after reconnect within 5 minutes with no data loss. Fail: Offline data is lost or requires manual re-entry.	Gardeners frequently work on rooftops with weak signal; the app must not lose their data.

Actors & Use Cases Extracted from Approved Requirements

Actors (3)

- Co-op Gardener — registers plots, logs crop data, posts tips, lists and trades surplus produce.
- Community Leader — reviews and approves plot registrations; can also browse listings.
- Community Member — browses and trades surplus produce listed by gardeners.

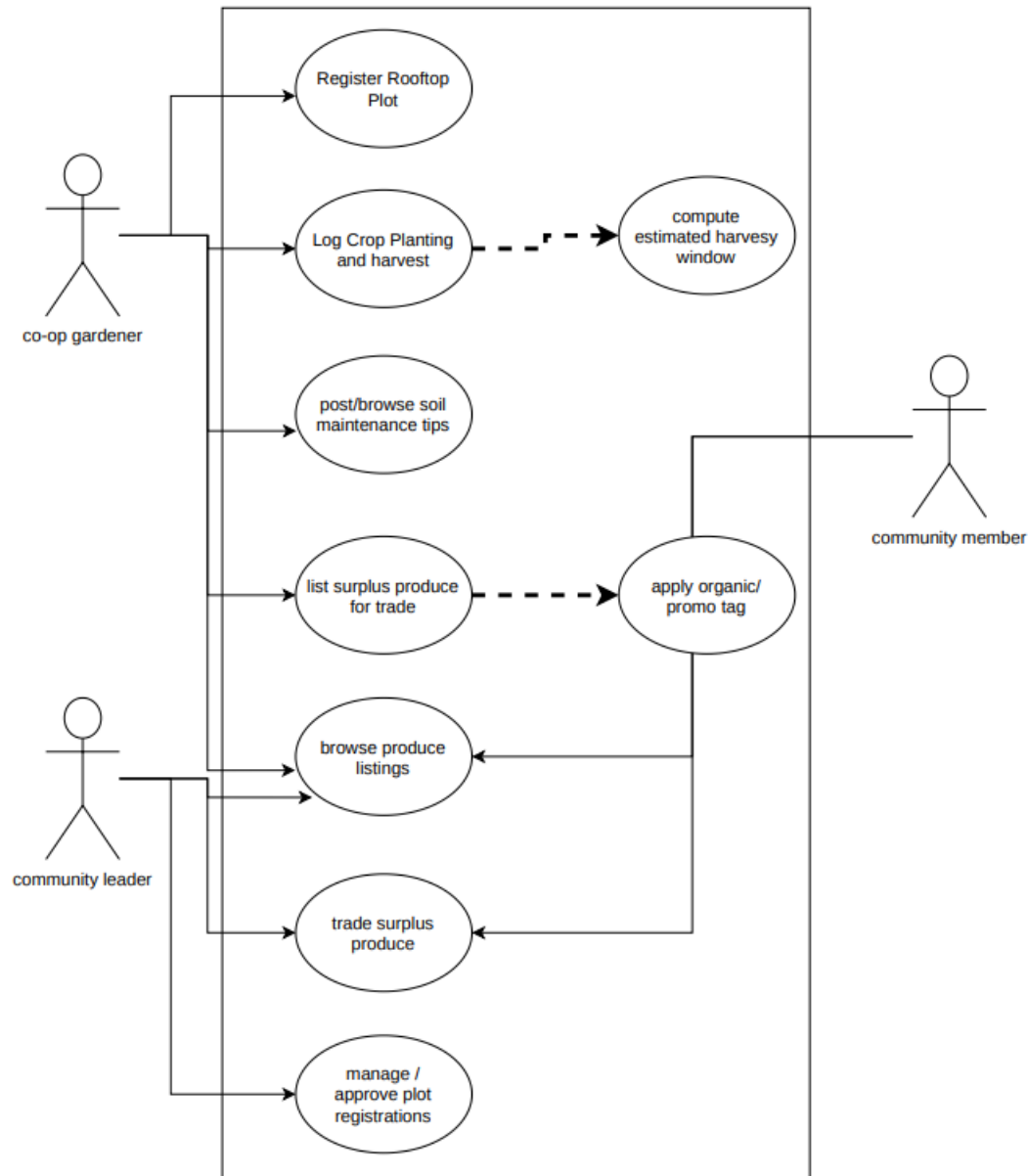
Use Cases (9)

1. Register Rooftop Plot — derived from FR-002.
2. Log Crop Planting and Harvest — derived from FR-001.
3. Compute Estimated Harvest Window — included by Use Case 2 (FR-001).
4. Post / Browse Soil Maintenance Tips — derived from FR-003.
5. List Surplus Produce for Trade — derived from FR-004.
6. Browse Produce Listings — supports FR-004 (Community Leader and Community Member).
7. Trade Surplus Produce — derived from FR-004 (Community Member side of the exchange).

8. Apply Organic / Promo Tag — extends Use Case 5 (FR-004), optional step.

9. Manage / Approve Plot Registrations — derived from FR-005.

Use Case Diagram



Use Case: List Surplus Produce for Trade

Primary Actor: Co-op Gardener

Related Requirements: FR-004 (list surplus produce), NFR-001 (real-time trade board updates)

Preconditions

- The gardener is a registered co-op member and is logged in to the app.
- The gardener has at least one approved rooftop plot with logged crop data.

Postconditions

- The produce listing is published and visible on the shared Trade Board to all co-op members.
- Nearby co-op members have been notified in real time of the new listing.

Main Success Scenario

1. Gardener selects “List Surplus Produce” from the app menu.
2. System displays a form requesting produce type, quantity, unit, and pickup window.
3. Gardener enters the produce details and submits the form.
4. System validates that the quantity is greater than zero and the pickup window is in the future.
5. Gardener may optionally apply an “Organic” or “Promo” tag to the listing (extension point).
6. System publishes the listing to the shared Trade Board.
7. System sends a real-time notification to nearby co-op members.
8. Use case ends successfully.

Alternate Flow

- 4a1. System displays an “Invalid quantity or pickup window” error message.
- 4a2. Gardener is prompted to correct the entry and resubmit.
- 4a3. If the gardener cancels instead of correcting the entry, the listing is discarded and the use case ends without publishing.